Homework 3

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Problem 1

Let $(F, +, \cdot)$ be a field.

(a) Prove that the multiplicative identity is unique: If $b \in F$ satisfies $b \cdot a = a \cdot b = a$ for all $a \in F$, then $b = 1$.

(b) Prove that multiplicative inverses are unique: If $a \in F \setminus \{0\}$ and $b \in F$ satisfy $b \cdot a = a \cdot b = 1$, then $b = a^{-1}$.

(c) Prove that if $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.

(d) Prove that if $a, b \in F$ then $-a + (-b) = -(a + b)$.

(e) Prove that if $a, b \in F$ and $a, b \neq 0$, then $a^{-1} \cdot b^{-1} = (ab)^{-1}$.

Discussion & Scratch Work

Main ideas (and why we cannot "just divide"). In a general field, "division" is not a primitive operation. Instead, we justify any division-like step by introducing multiplicative inverses via **(M5)** and then using associativity **(M3)** and the identity **(M4)**.

- **(a)** Compare $b \cdot 1$ computed from the hypothesis and from **(M4)**.
- **(b)** Insert $1 = a \cdot a^{-1}$ (from **(M5)**) and reassociate (by **(M3)**) to force $b = a^{-1}$.
- **(c)** Prove cancellation by multiplying $a \cdot b = a \cdot c$ on the left by a^{-1} (existence from **(M5)**), then use **(M3)** and **(M4)**.
- **(d)** Show $(-a) + (-b)$ is an additive inverse of $(a + b)$ by checking $(a + b) + ((-a) + (-b)) = 0$ using **(A2)**, **(A3)**, **(A4)**, **(A5)**. Conclude it equals $-(a + b)$ by uniqueness of additive inverses (proved from the axioms, as in lecture).
- **(e)** Show $a^{-1} \cdot b^{-1}$ is an inverse of ab using commutativity **(M2)**, associativity **(M3)**, and **(M5)**. Then apply (b) to conclude $a^{-1}b^{-1} = (ab)^{-1}$.

Proof.
Let $(F, +, \cdot)$ be a field.

(a) Suppose $b \in F$ satisfies

$$b \cdot a = a \cdot b = a \quad \text{for all } a \in F.$$

Let $1 \in F$ be the multiplicative identity whose existence is guaranteed by **(M4)**. Setting $a = 1$ in the hypothesis gives

$$b \cdot 1 = 1.$$

But by **(M4)** (applied with $a = b$), we also have $b \cdot 1 = b$. Therefore $b = 1$, so the multiplicative identity is unique.

(b) Suppose $a \in F \setminus \{0\}$ and $b \in F$ satisfy

$$b \cdot a = a \cdot b = 1.$$

By the multiplicative inverse axiom **(M5)**, there exists $a^{-1} \in F$ such that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

We prove that $b = a^{-1}$. Using the multiplicative identity axiom **(M4)**, we have

$$b = b \cdot 1.$$

Substituting $1 = a \cdot a^{-1}$ from **(M5)** gives

$$b = b \cdot (a \cdot a^{-1}).$$

By associativity of multiplication **(M3)**,

$$b \cdot (a \cdot a^{-1}) = (b \cdot a) \cdot a^{-1}.$$

Using the hypothesis $b \cdot a = 1$, this becomes

$$(b \cdot a) \cdot a^{-1} = 1 \cdot a^{-1}.$$

Finally, by **(M4)** we have $1 \cdot a^{-1} = a^{-1}$. Therefore $b = a^{-1}$, so the multiplicative inverse of a is unique.

(c) Suppose $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$. By **(M5)**, there exists $a^{-1} \in F$ such that $a^{-1} \cdot a = 1$. Multiply the equality $a \cdot b = a \cdot c$ on the left by a^{-1}:

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot (a \cdot c).$$

By associativity **(M3)**,

$$(a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c.$$

Using $a^{-1} \cdot a = 1$ (from **(M5)**) and the identity axiom **(M4)**, we get

$$1 \cdot b = 1 \cdot c \quad \Rightarrow \quad b = c.$$

(d) We claim that $-a + (-b)$ is the additive inverse of $a + b$, hence equals $-(a + b)$. First compute, using associativity **(A3)** and commutativity **(A2)** of $+$,

$$\begin{aligned}
(a+b) + \big((-a) + (-b)\big) &= a + \Big(b + \big((-a) + (-b)\big)\Big) \quad \text{by } \textbf{(A3)} \\
&= a + \Big(\big(b + (-b)\big) + (-a)\Big) \quad \text{by } \textbf{(A3)} \text{ and } \textbf{(A2)} \\
&= a + \big(0 + (-a)\big) \quad \text{by } \textbf{(A5)} \\
&= a + (-a) \quad \text{by } \textbf{(A4)} \\
&= 0 \quad \text{by } \textbf{(A5)}.
\end{aligned}$$

Thus $(-a) + (-b)$ is an additive inverse of $a + b$.

Uniqueness of additive inverses (from axioms). If $x, y \in F$ both satisfy $(a + b) + x = 0$ and $(a+b)+y = 0$, then adding $-(a+b)$ to both sides and using **(A3)**, **(A4)**, **(A5)** yields $x = y$ (as in lecture). Therefore the additive inverse of $a + b$ is unique, and since $-(a + b)$ denotes that unique inverse, we conclude

$$-a + (-b) = (-a) + (-b) = -(a + b).$$

(e) Assume $a, b \in F$ with $a \neq 0$ and $b \neq 0$. By **(M5)**, there exist $a^{-1}, b^{-1} \in F$ such that $a \cdot a^{-1} = 1$ and $b \cdot b^{-1} = 1$. Consider

$$(ab) \cdot (a^{-1} \cdot b^{-1}).$$

By associativity **(M3)** and commutativity **(M2)**,

$$(ab) \cdot (a^{-1}b^{-1}) = a \cdot b \cdot a^{-1} \cdot b^{-1} = (a \cdot a^{-1}) \cdot (b \cdot b^{-1}).$$

Using **(M5)** and then **(M4)**, this equals $1 \cdot 1 = 1$. Similarly,

$$(a^{-1}b^{-1}) \cdot (ab) = 1.$$

Hence $a^{-1}b^{-1}$ is a multiplicative inverse of ab. By part **(b)** (uniqueness of inverses),

$$a^{-1} \cdot b^{-1} = (ab)^{-1}.$$

$\square$

Problem 2

Given two points (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$, we write

$$(x_1, y_1) < (x_2, y_2) \Leftrightarrow x_1 < x_2 \text{ or } (x_1 = x_2 \text{ and } y_1 < y_2).$$

(a) Prove that $<$ is an order relation on $\mathbb{R}^2$.

(b) Prove that $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property. (Hint: Consider the set of all points on the y-axis.)

Discussion & Scratch Work

Main ideas. This is the *lexicographic order* on $\mathbb{R}^2$.

- **(a)** To prove $<$ is an order relation on $\mathbb{R}^2$, verify the two defining properties: **trichotomy** and **transitivity**. The proofs reduce to the corresponding properties of the usual order on $\mathbb{R}$ by splitting into cases on the first coordinates.
- **(b)** To show the least upper bound property fails, use the set $A = \{(0, y) : y \in \mathbb{R}\}$ (the y-axis). It is bounded above (e.g. by $(1, 0)$), but any upper bound must have first coordinate > 0, and then you can always find a strictly smaller upper bound by decreasing the first coordinate (e.g. replace (x, y) by $(x/2, 0)$). Hence no least upper bound exists.

Proof.
Let $p = (x_1, y_1)$, $q = (x_2, y_2)$, and $r = (x_3, y_3)$ be points in $\mathbb{R}^2$.

(a) We prove that $<$ is an order relation on $\mathbb{R}^2$ in the sense of the lecture definition.

Trichotomy. By trichotomy of the usual order on $\mathbb{R}$, exactly one of $x_1 < x_2$, $x_1 = x_2$, or $x_2 < x_1$ holds.

- If $x_1 < x_2$, then $p < q$ by definition.
- If $x_2 < x_1$, then $q < p$ by definition.
- If $x_1 = x_2$, then by trichotomy on $\mathbb{R}$ applied to y_1, y_2, exactly one of $y_1 < y_2$, $y_1 = y_2$, or $y_2 < y_1$ holds. In the first case, $p < q$; in the second, $p = q$; in the third, $q < p$.

Therefore exactly one of $p < q$, $p = q$, or $q < p$ holds.

Transitivity. Assume $p < q$ and $q < r$. We show $p < r$.

If $x_1 < x_2$, then from $q < r$ we have either $x_2 < x_3$ or $x_2 = x_3$. In both cases $x_2 \leq x_3$, hence $x_1 < x_3$, so $p < r$ by definition.

If instead $x_1 = x_2$, then $p < q$ forces $y_1 < y_2$. Now consider $q < r$:

- If $x_2 < x_3$, then $x_1 = x_2 < x_3$, so $p < r$.

- If $x_2 = x_3$, then $q < r$ forces $y_2 < y_3$, and by transitivity in $\mathbb{R}$ we get $y_1 < y_3$, so (since $x_1 = x_3$) we conclude $p < r$.

Thus $<$ is transitive.

Since $<$ satisfies trichotomy and transitivity, it is an order relation on $\mathbb{R}^2$.

(b) Let

$$A := \{(0, y) : y \in \mathbb{R}\} \subseteq \mathbb{R}^2.$$

Then A is nonempty (for example $(0, 0) \in A$). We claim A is bounded above in $(\mathbb{R}^2, <)$. Indeed, $(1, 0)$ is an upper bound, because for any $(0, y) \in A$ we have $0 < 1$, hence $(0, y) < (1, 0)$ by the definition of $<$.

We show A has *no* least upper bound. First note that if (u, v) is any upper bound of A, then necessarily $u > 0$. If $u < 0$, then $(0, 0) \in A$ would satisfy $(u, v) < (0, 0)$, so (u, v) could not be an upper bound. If $u = 0$, then being an upper bound would require $y \leq v$ for all $y \in \mathbb{R}$, which is impossible since A contains all points $(0, y)$ with $y \in \mathbb{R}$ and $\mathbb{R}$ is unbounded above.

Now take any upper bound (u, v) of A; then $u > 0$. Define $w := (u/2, 0)$. For any $(0, y) \in A$ we have $0 < u/2$, hence $(0, y) < (u/2, 0) = w$, so w is also an upper bound of A. But $w < (u, v)$ because $u/2 < u$.

Therefore every upper bound of A is strictly larger than some other upper bound of A, so no least upper bound exists. Hence $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property. $\square$

Problem 3

Recall that the set of complex numbers $\mathbb{C}$ consists of all numbers of the form $a + ib$ for $a, b \in \mathbb{R}$. We can add and multiply complex numbers as follows:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \quad \text{and} \quad (a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

With these operations, it turns out $(\mathbb{C}, +, \cdot)$ is a field (but you do not need to prove this).

(a) The definition of a field says that for any $a + ib \neq 0 + i0$, there exists $c + id \in \mathbb{C}$ so that $(a + ib) \cdot (c + id) = 1 + i0$. What are $c, d \in \mathbb{R}$ in terms of $a, b \in \mathbb{R}$? Prove your answer.

(b) Prove that there is no order relation $<$ on $\mathbb{C}$ that makes $(\mathbb{C}, +, \cdot, <)$ an ordered field. (Hint: Can $i = 0 + i1$ be positive? Negative? Zero?)

Discussion & Scratch Work

Main ideas.

- **(a)** To find $(a+ib)^{-1}$, write an unknown inverse as $c+id$ and expand $(a+ib)(c+id) = (ac - bd) + i(ad + bc)$. Matching real and imaginary parts with $1 + i0$ gives a 2×2 linear system. Solve it to get

$$c = \frac{a}{a^2 + b^2}, \qquad d = -\frac{b}{a^2 + b^2}.$$

  The key point is that $a^2 + b^2 \neq 0$ whenever $a + ib \neq 0$.

- **(b)** In an ordered field, we always have $1 > 0$ and $x \neq 0 \Rightarrow x^2 > 0$ (Appendix, **(P3)**). But in $\mathbb{C}$ we have $i^2 = -1$. Trichotomy forces $i > 0$, $i = 0$, or $i < 0$; each case contradicts $x^2 > 0$ together with $1 > 0$. So no order can make $\mathbb{C}$ into an ordered field.

Proof.
(a) Let $a + ib \in \mathbb{C}$ with $a + ib \neq 0 + i0$. We seek $c + id \in \mathbb{C}$ such that

$$(a + ib)(c + id) = 1 + i0.$$

Using the given multiplication rule,

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc).$$

Thus we require

$$ac - bd = 1, \qquad ad + bc = 0. \tag{1}$$

Multiply the first equation in (1) by a and the second by b, then add:

$$a(ac - bd) + b(ad + bc) = a.$$

The left-hand side simplifies to $(a^2 + b^2)c$, so

$$(a^2 + b^2)c = a.$$

Similarly, multiply the second equation by a and the first by b, then subtract:

$$a(ad + bc) - b(ac - bd) = 0 - b,$$

whose left-hand side simplifies to $(a^2 + b^2)d$, so

$$(a^2 + b^2)d = -b.$$

Because $a + ib \neq 0$, we have $(a, b) \neq (0, 0)$, hence $a^2 + b^2 > 0$ (since $a^2 \geq 0$ and $b^2 \geq 0$, and not both are zero, we have $a^2 + b^2 > 0$) and in particular $a^2 + b^2 \neq 0$. Therefore we may divide by $a^2 + b^2$ to obtain

$$c = \frac{a}{a^2 + b^2}, \qquad d = -\frac{b}{a^2 + b^2}.$$

A direct substitution into (1) (or into $(ac - bd) + i(ad + bc)$) shows that these values indeed yield $(a+ib)(c+id) = 1+i0$. This proves the claimed formula for the multiplicative inverse.

(b) Suppose, for contradiction, that there exists an order relation $<$ on $\mathbb{C}$ such that $(\mathbb{C}, +, \cdot, <)$ is an ordered field. By Appendix **(P3)**, we have $1 > 0$ and, for any $x \in \mathbb{C}$, $x \neq 0 \Rightarrow x^2 > 0$. By trichotomy, exactly one of $i > 0$, $i = 0$, or $i < 0$ holds. We treat each case:

- If $i = 0$, then $i^2 = 0$, contradicting $i^2 = -1$.
- If $i > 0$, then $i \neq 0$, so Appendix **(P3)** implies $i^2 > 0$. But $i^2 = -1$, so $-1 > 0$, hence $1 < 0$, contradicting $1 > 0$.
- If $i < 0$, then $-i > 0$ (using Appendix **(P1)** with $a = i$). Since $-i \neq 0$, Appendix **(P3)** gives $(-i)^2 > 0$. But $(-i)^2 = i^2 = -1$, leading again to $-1 > 0$ and hence $1 < 0$, a contradiction.

All cases contradict the ordered-field consequences, so no such order relation exists. $\square$

Problem 4

Let $A \subseteq \mathbb{R}$ be a nonempty and bounded set, let $c \in \mathbb{R}$ be a number, and define

$$B = \{c \cdot a : a \in A\}.$$

(a) Prove that if $c > 0$, then $\sup B = c \cdot \sup A$.

(b) Prove that if $c < 0$, then $\sup B = c \cdot \inf A$.

Discussion & Scratch Work

> **Main idea.** We prove an equality of suprema by showing the candidate value is (i) an upper bound for B and (ii) the *least* such upper bound. The only subtlety is that we may multiply inequalities by a positive number without changing the direction (ordered-field axiom **(O2)**), while multiplying by a negative number reverses the inequality (Appendix **(P2)**).
>
> - If $c > 0$, then $a \leq \sup A$ implies $ca \leq c \sup A$, so $c \sup A$ is an upper bound for B. For leastness, any upper bound M of B yields $ca \leq M$ for all $a \in A$, hence $a \leq M/c$ after multiplying by $c^{-1} > 0$, so M/c is an upper bound for A and $\sup A \leq M/c$.
> - If $c < 0$, then $\inf A \leq a$ implies $c \inf A \geq ca$, so $c \inf A$ is an upper bound for B. For leastness, if $ca \leq M$ for all $a \in A$, multiplying by $c^{-1} < 0$ reverses the inequality and gives $a \geq M/c$, so M/c is a lower bound for A and hence $M/c \leq \inf A$.

Proof.
Let $A \subseteq \mathbb{R}$ be nonempty and bounded, let $c \in \mathbb{R}$, and set $B = \{ca : a \in A\}$. We use the definitions of sup and inf from the Appendix.

(a) Assume $c > 0$. We prove $\sup B = c \sup A$.

Step 1: $c \sup A$ is an upper bound for B. Let $b \in B$. Then $b = ca$ for some $a \in A$. Because $\sup A$ is an upper bound for A, we have $a \leq \sup A$. Since $c > 0$, multiplying by c preserves the inequality (ordered-field axiom **(O2)**), so

$$ca \leq c \sup A.$$

Thus $b \leq c \sup A$, and $c \sup A$ is an upper bound for B.

Step 2: $c \sup A$ is the least upper bound for B. Let $M \in \mathbb{R}$ be any upper bound for B. Then for every $a \in A$, we have $ca \in B$, hence

$$ca \leq M.$$

Because $c > 0$, we have $c \neq 0$ and $c^{-1} > 0$. Multiplying by c^{-1} preserves the inequality (**(O2)**), giving

$$a \leq c^{-1}M = \frac{M}{c} \quad \text{for all } a \in A.$$

So M/c is an upper bound for A. By definition of $\sup A$, this implies $\sup A \leq M/c$. Multiplying by $c > 0$ again preserves the inequality (**(O2)**), so

$$c \sup A \leq M.$$

Therefore $c \sup A$ is the least upper bound of B, i.e. $\sup B = c \sup A$.

(b) Assume $c < 0$. We prove $\sup B = c \inf A$.

Step 1: $c \inf A$ is an upper bound for B. Let $b \in B$, so $b = ca$ for some $a \in A$. Because $\inf A$ is a lower bound for A, we have $\inf A \leq a$. Since $c < 0$, multiplying by c reverses the inequality (Appendix **(P2)**), hence

$$c \inf A \geq ca.$$

Equivalently, $ca \leq c \inf A$, so $b \leq c \inf A$. Thus $c \inf A$ is an upper bound for B.

Step 2: $c \inf A$ is the least upper bound for B. Let $M \in \mathbb{R}$ be any upper bound for B. Then for every $a \in A$,

$$ca \leq M.$$

Because $c < 0$, we have $c \neq 0$ and $c^{-1} < 0$. Multiplying by c^{-1} reverses the inequality (Appendix **(P2)**), so

$$a \geq c^{-1}M = \frac{M}{c} \quad \text{for all } a \in A.$$

Thus M/c is a lower bound for A. By definition of $\inf A$ (greatest lower bound), we get $M/c \leq \inf A$. Multiplying by $c < 0$ reverses the inequality again (Appendix **(P2)**), yielding

$$M \geq c \inf A.$$

So every upper bound M of B satisfies $c \inf A \leq M$, hence $c \inf A$ is the least upper bound of B. Therefore $\sup B = c \inf A$. $\square$

Problem 5

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that $\sup S = \sup A + \sup B$.

Discussion & Scratch Work

Main idea. Let $\alpha := \sup A$ and $\beta := \sup B$. We show

$$\sup S = \alpha + \beta$$

by the standard two-step pattern for suprema:

- **Upper bound:** For any $s \in S$, write $s = a + b$ with $a \in A$, $b \in B$. Since $a \leq \alpha$ and $b \leq \beta$, we get $s \leq \alpha + \beta$. Hence $\alpha + \beta$ is an upper bound for S.
- **Leastness:** If M is any upper bound for S, then for each fixed $b \in B$ we have $a + b \leq M$ for all $a \in A$, hence $a \leq M - b$. Thus $M - b$ is an upper bound for A, so $\alpha \leq M - b$, i.e. $\alpha + b \leq M$ for all $b \in B$. Therefore $M - \alpha$ is an upper bound for B, so $\beta \leq M - \alpha$, which gives $\alpha + \beta \leq M$.

This proves $\alpha + \beta$ is the least upper bound of S.

Proof.
Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and set

$$S := \{a + b : a \in A \text{ and } b \in B\}.$$

Let $\alpha := \sup A$ and $\beta := \sup B$.

Let $s \in S$. Then there exist $a \in A$ and $b \in B$ with $s = a + b$. Since α is an upper bound for A and β is an upper bound for B, we have $a \leq \alpha$ and $b \leq \beta$. Adding these inequalities gives

$$a + b \leq \alpha + \beta,$$

so $s \leq \alpha + \beta$. Hence $\alpha + \beta$ is an upper bound for S.

Let $M \in \mathbb{R}$ be any upper bound for S. Fix $b \in B$. For every $a \in A$, we have $a + b \in S$, hence $a + b \leq M$. Adding $-b$ to both sides (by **(O1)**) yields $a \leq M - b$ for all $a \in A$, so $M - b$ is an upper bound for A. By definition of $\alpha = \sup A$, we get

$$\alpha \leq M - b,$$

which implies $\alpha + b \leq M$. Because this holds for every $b \in B$, we have $b \leq M - \alpha$ for all $b \in B$, so $M - \alpha$ is an upper bound for B. By definition of $\beta = \sup B$, we obtain

$$\beta \leq M - \alpha.$$

Adding α to both sides gives $\alpha + \beta \leq M$.

Since $\alpha + \beta$ is an upper bound for S and is $\leq$ every upper bound M of S, it is the least upper bound. Therefore

$$\sup S = \alpha + \beta = \sup A + \sup B.$$

$\square$

Problem 6

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets. Prove or disprove the following statements:

(a) If $a \leq b$ for all $a \in A$ and for all $b \in B$, then $\sup A \leq \inf B$.

(b) If $a < b$ for all $a \in A$ and for all $b \in B$, then $\sup A < \inf B$.

Discussion & Scratch Work

Main ideas. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded.

- **(a) True.** The hypothesis

$$a \leq b \quad \text{for all } a \in A \text{ and all } b \in B$$

says: *every element of B is an upper bound for A.* By the definition of supremum, $\sup A$ is *below every* upper bound of A, hence $\sup A \leq b$ for all $b \in B$. That means $\sup A$ is a *lower bound* for B, so by the definition of infimum (greatest lower bound), $\sup A \leq \inf B$.

- **(b) False in general.** Even if $a < b$ for all $a \in A$ and $b \in B$, it can still happen that $\sup A = \inf B$ because $\sup A$ need not belong to A. A clean counterexample is

$$A = (0, 1), \qquad B = \{1\}.$$

Then $a < 1$ for all $a \in (0, 1)$, but $\sup A = 1$ and $\inf B = 1$.

Proof.
Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded.

(a) Assume that $a \leq b$ for all $a \in A$ and all $b \in B$. We prove that $\sup A \leq \inf B$.

Fix any $b \in B$. By the hypothesis, $a \leq b$ for every $a \in A$, so b is an upper bound for A (Appendix, Definition **(B1)**). Let $\alpha := \sup A$. By the defining property of the supremum (Appendix, Definition **(S1)**), we have

$$\alpha \leq b.$$

Because $b \in B$ was arbitrary, we conclude that $\alpha \leq b$ for all $b \in B$, i.e. α is a lower bound for B (Appendix, Definition **(B2)**).

Let $\beta := \inf B$. By the defining property of the infimum (Appendix, Definition **(S2)**), β is the *greatest* lower bound of B, so every lower bound of B is $\leq \beta$. In particular, since α is a lower bound of B, we obtain

$$\sup A = \alpha \leq \beta = \inf B.$$

(b) The statement is *false*. Consider

$$A := (0, 1), \qquad B := \{1\}.$$

Then A and B are nonempty and bounded subsets of $\mathbb{R}$. Moreover, for every $a \in A$ and every $b \in B$ we have $b = 1$ and $0 < a < 1$, hence $a < b$.

However, $\sup A = 1$ (because 1 is an upper bound for $(0, 1)$ and any smaller number fails to be an upper bound), and clearly $\inf B = 1$ because B consists of the single element 1. Therefore

$$\sup A = 1 = \inf B,$$

so it is *not* true that $\sup A < \inf B$ in general. $\square$

Appendix: Toolbox

Basically a summary of lectures 6—9; something like a little cheatsheet review. Anyways, below is the good stuff.

Field axioms

(A1) **(Closure for $+$)** If $a, b \in F$, then $a + b \in F$.

(A2) **(Commutativity of $+$)** If $a, b \in F$, then $a + b = b + a$.

(A3) **(Associativity of $+$)** If $a, b, c \in F$, then $(a + b) + c = a + (b + c)$.

(A4) **(Additive identity)** There exists $0 \in F$ such that $a + 0 = 0 + a = a$ for all $a \in F$.

(A5) **(Additive inverses)** For each $a \in F$ there exists $-a \in F$ such that $a + (-a) = (-a) + a = 0$.

(M1) **(Closure for $\cdot$)** If $a, b \in F$, then $a \cdot b \in F$.

(M2) **(Commutativity of $\cdot$)** If $a, b \in F$, then $a \cdot b = b \cdot a$.

(M3) **(Associativity of $\cdot$)** If $a, b, c \in F$, then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

(M4) **(Multiplicative identity)** There exists $1 \in F$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in F$.

(M5) **(Multiplicative inverses)** For each $a \in F \setminus \{0\}$ there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

(D) **(Distributivity)** If $a, b, c \in F$, then $(a + b) \cdot c = a \cdot c + b \cdot c$.

Derived properties of fields (proved from the axioms in Lecture 7)

(1) The additive and multiplicative identities are unique.

(2) Additive and multiplicative inverses are unique.

(3) If $a, b, c \in F$ satisfy $a + b = a + c$, then $b = c$. If $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.

(4) For all $a \in F$, we have $a \cdot 0 = 0 \cdot a = 0$.

(5) For all $a, b \in F$, we have $(-a) \cdot b = a \cdot (-b) = -(a \cdot b)$.

(6) For all $a, b \in F$, we have $(-a) \cdot (-b) = a \cdot b$.

(7) If $a \cdot b = 0$, then $a = 0$ or $b = 0$.

Order relations and ordered fields (Lectures 7–8)

Order relations

Definition (Order relation). Let A be a nonempty set. An *order relation* (or *inequality*) on A is a relation $<$ on A such that:

(OR1) **(Trichotomy)** If $a, b \in A$, then exactly one of the following holds: $a < b$, or $a = b$, or $b < a$.

(OR2) **(Transitivity)** If $a, b, c \in A$ satisfy $a < b$ and $b < c$, then $a < c$.

Notation.

- $a > b$ means $b < a$.
- $a \leq b$ means $(a < b)$ or $(a = b)$.
- $a \geq b$ means $(b < a)$ or $(a = b)$.

Ordered fields

Definition (Ordered field). Let $(F, +, \cdot)$ be a field. We say that $(F, +, \cdot, <)$ is an *ordered field* if it is equipped with an order relation $<$ on F such that:

(O1) If $a, b, c \in F$ satisfy $a < b$, then $a + c < b + c$.

(O2) If $a, b, c \in F$ satisfy $a < b$ and $0 < c$, then $a \cdot c < b \cdot c$.

Useful consequences (proved in Lecture 8). Let $(F, +, \cdot, <)$ be an ordered field.

(P1) $a > 0 \iff -a < 0$.

(P2) If $a, b, c \in F$ satisfy $a < b$ and $c < 0$, then $a \cdot c > b \cdot c$.

(P3) If $a \in F$ satisfies $a \neq 0$, then $a^2 = a \cdot a > 0$. In particular, $1 > 0$.

(P4) If $a, b \in F$ satisfy $0 < a < b$, then $0 < b^{-1} < a^{-1}$.

Bounds, maxima/minima, and suprema/infima (Lecture 8)

Bounds and extrema

Definition (Bounds; max/min). Let $(F, +, \cdot, <)$ be an ordered field, and let $A \subseteq F$ be nonempty.

(B1) A is *bounded above* if there exists $M \in F$ such that $a \leq M$ for all $a \in A$. Any such M is an *upper bound* for A. If, in addition, $M \in A$, then M is the *maximum* of A.

(B2) A is *bounded below* if there exists $m \in F$ such that $m \leq a$ for all $a \in A$. Any such m is a *lower bound* for A. If, in addition, $m \in A$, then m is the *minimum* of A.

(B3) A is *bounded* if it is both bounded above and bounded below.

Remark (Max/min need not exist; uniqueness). Some sets do not have a maximum (or a minimum). If a maximum exists, then it is unique: if $M, M' \in A$ both satisfy $a \leq M$ for all $a \in A$ and $a \leq M'$ for all $a \in A$, then $M \leq M'$ and $M' \leq M$, hence $M = M'$. (The same statement holds for minima.)

Remark. Even if a set does not have a maximum, it may still have (at least one) upper bound.

Least upper bounds and greatest lower bounds

Definition (Supremum/infimum). Let $(F, +, \cdot, <)$ be an ordered field, and let $A \subseteq F$ be nonempty.

(S1) A number $L \in F$ is the *least upper bound* (or *supremum*) of A if:

- L is an upper bound for A, and
- if M is an upper bound for A, then $L \leq M$.

When this is true, we write $L = \sup A$.

(S2) A number $\ell \in F$ is the *greatest lower bound* (or *infimum*) of A if:

- ℓ is a lower bound for A, and
- if m is a lower bound for A, then $m \leq \ell$.

When this is true, we write $\ell = \inf A$.

Lemma (Uniqueness of the supremum). If $A \subseteq F$ has a least upper bound, then it is unique.

Proof. Suppose $L, M \in F$ are both least upper bounds for A. Then L is an upper bound and, since M is a least upper bound, we have $M \leq L$. Similarly, M is an upper bound and, since L is a least upper bound, we have $L \leq M$. Therefore $L = M$. $\square$

Completeness and the real numbers (Lectures 8–9)

Proposition 1 (Irrationality of $\sqrt{2}$)**.** *We have* $\sqrt{2} \notin \mathbb{Q}$.

Proof. Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we may write

$$\sqrt{2} = \frac{a}{b}$$

for some integers $a \in \mathbb{Z}$ and $b \in \mathbb{N}$ with no common factors (i.e. the fraction is in lowest terms). Squaring gives

$$2 = \frac{a^2}{b^2} \qquad \Rightarrow \qquad 2b^2 = a^2.$$

Thus a^2 is even, hence a is even. Write $a = 2k$ for some $k \in \mathbb{Z}$. Substituting yields

$$2b^2 = (2k)^2 = 4k^2 \qquad \Rightarrow \qquad b^2 = 2k^2.$$

So b^2 is even, hence b is even. Therefore both a and b are even, contradicting that a/b was in lowest terms. Hence $\sqrt{2} \notin \mathbb{Q}$. □

Definition 1 (Least upper bound property)**.** *Let $<$ be an order relation on a set F. We say that F has the* least upper bound property *if for every nonempty subset $A \subseteq F$ that is bounded above, there exists a least upper bound of A that belongs to F (i.e. $\sup A \in F$).*

Remark. *The ordered field $(\mathbb{Q}, +, \cdot, <)$ does* not *have the least upper bound property. For example,*

$$A := \{x \in \mathbb{Q} : x^2 < 2\}$$

is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum in $\mathbb{Q}$ (its "expected" supremum would be $\sqrt{2}$, which is not rational).

Theorem 1 (Existence and uniqueness of $\mathbb{R}$ as a complete ordered field)**.** *There exists an ordered field with the least upper bound property. We denote it by*

$$(\mathbb{R}, +, \cdot, <).$$

This ordered field is unique up to order-preserving field isomorphism, and moreover $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Proposition 2 (Existence of infima in $\mathbb{R}$)**.** *If $A \subseteq \mathbb{R}$ is nonempty and bounded below, then $\inf A$ exists in $\mathbb{R}$.*

Proof. Assume $A \subseteq \mathbb{R}$ is nonempty and bounded below. Define

$$-A := \{-a : a \in A\}.$$

Then $-A$ is nonempty, and $-A$ is bounded above (if m is a lower bound for A, then $-m$ is an upper bound for $-A$). By the least upper bound property of $\mathbb{R}$, $\sup(-A)$ exists in $\mathbb{R}$. We claim that

$$\inf A = -\sup(-A).$$

Let $L := \sup(-A)$. For any $a \in A$ we have $-a \in -A$, hence $-a \leq L$, i.e. $-L \leq a$. So $-L$ is a lower bound for A. If m is any lower bound for A, then $m \leq a$ for all $a \in A$, so $-a \leq -m$ for all $a \in A$, meaning $-m$ is an upper bound for $-A$. By definition of $L = \sup(-A)$, we have $L \leq -m$, hence $m \leq -L$. Thus $-L$ is the greatest lower bound of A, so $\inf A = -L = -\sup(-A)$. □

Remark (A useful identity)**.** *For any nonempty $A \subseteq \mathbb{R}$ that is bounded below,*

$$\inf A = -\sup(-A), \qquad \textit{where } -A := \{-a : a \in A\}.$$

Conceptually: turning a lower-bound problem for A into an upper-bound problem for $-A$ lets us apply completeness (the LUB property).